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Towards Assumption-Based Argumentation Mining in Hotel Reviews

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INTRODUCTION

This project applies the Assumption-Based Argumentation (ABA) framework to hotel reviews, focusing on the task of classifying topics and sentiments as a foundation for mining arguments from unstructured text. We evaluate large language models (LLMs), including GPT-4o, Gemini 2.5 Pro, LLaMA 3, and Mistral, using prompt templates designed across zero- to five-shot configurations on a custom-annotated dataset. Our evaluation, based on topic match accuracy, sentiment accuracy, and cosine similarity, shows that carefully engineered prompts combined with lower temperature settings can yield strong performance. At the same time, we identify key challenges such as increased execution time at higher shot sizes and inconsistent output formatting, particularly in some models. This study provides an initial benchmark for leveraging LLMs in ABA-based argument mining and points to areas for future improvement.

WHAT IS ABA?

Assumption-Based Argumentation (ABA) is a logic-based framework for structuring reasoning. It begins with assumptions and uses inference rules to draw conclusions, while contraries define what can be attacked. Arguments are then built from assumptions, and attacks occur when one argument contradicts another. In the context of hotel reviews, ABA helps transform unstructured opinions into structured arguments, revealing both supporting and opposing viewpoints.

TECHNICAL TOOLS

Gemini

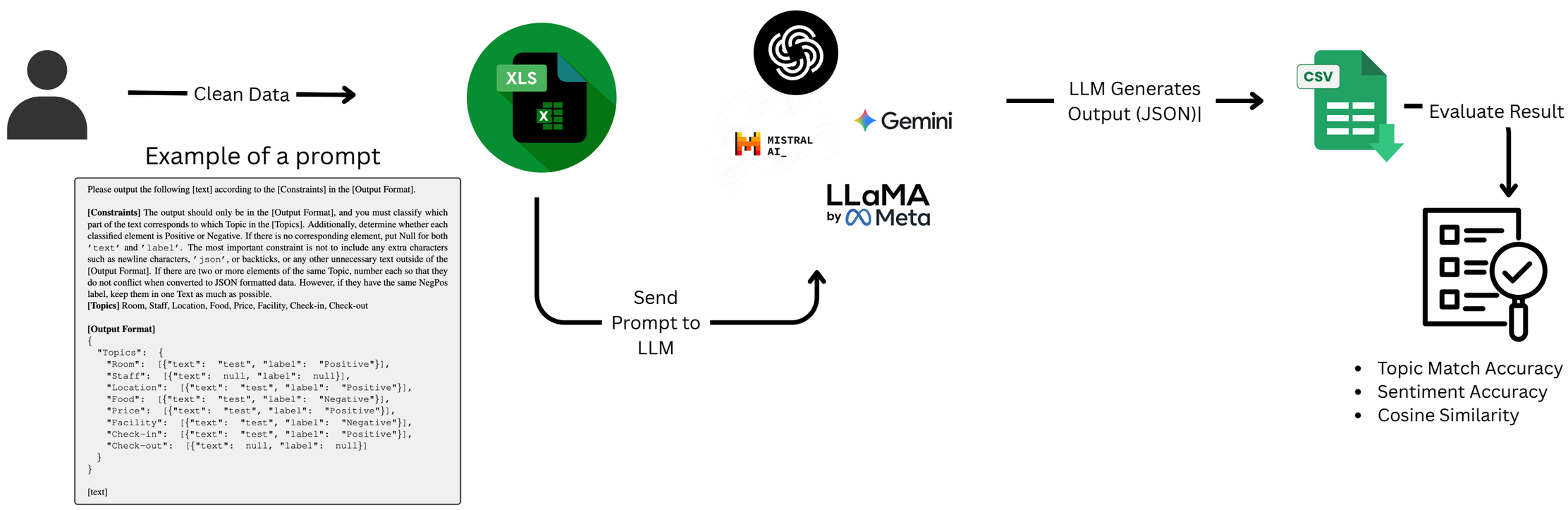
MISTRAL AI_

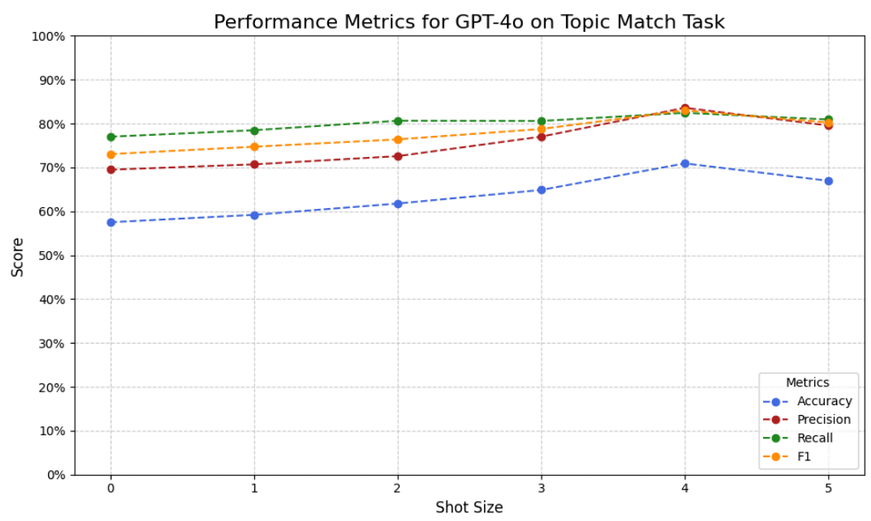
LLaMA by Meta

SCOPE OF WORK

- **In-Context Learning Setup:** Apply zero- to five-shot configurations to explore how different amounts of examples influence LLM performance.
- LLM Evaluation: Benchmark GPT-4o, Gemini 2.5 Pro, LLaMA 3, and Mistral on topic and sentiment classification tasks.Evaluation Metrics: Measure performance using topic match accuracy, sentiment accuracy, and cosine similarity.
- ABA Foundation: Establish a baseline for linking LLM-based topic and sentiment classification to the broader Assumption-Based Argumentation (ABA) framework.

PIPELINE OF WORK



EXAMPLE OF RESULT

ID	Topics	Text	NegPos
1	Room	New, comfortable a	Positive
1	Staff	Staff is extremely he	Positive
1	Location	close to the airport,	Positive
1	Food	Tasty food on the fir	Positive
1	Price		
1	Facility		
1	Check-in		
1	Check-out